Operation Guide for SDK Configuration of AEC Support in the Xiaozhi AI Terminal Open Source Project

This document is intended only for developers, at least beginners who have already started

Modification Premise:

1) Please first ensure that the ESP-IDF Development Environment has been installed and configured, with a version above IDF 5.0 (each major version of IDF is incompatible). To avoid unnecessary issues, it is recommended to use **IDF V5.3.2**.

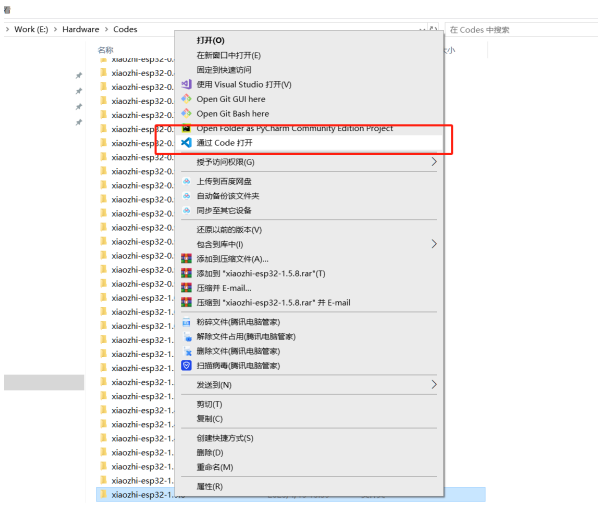
For the installation and configuration of the ESP-IDF Development Environment on Windows 10, please refer to the previous tutorial document: Setting up the ESP IDF 5.3.2 Development Environment on Windows and Compiling Xiaozhi

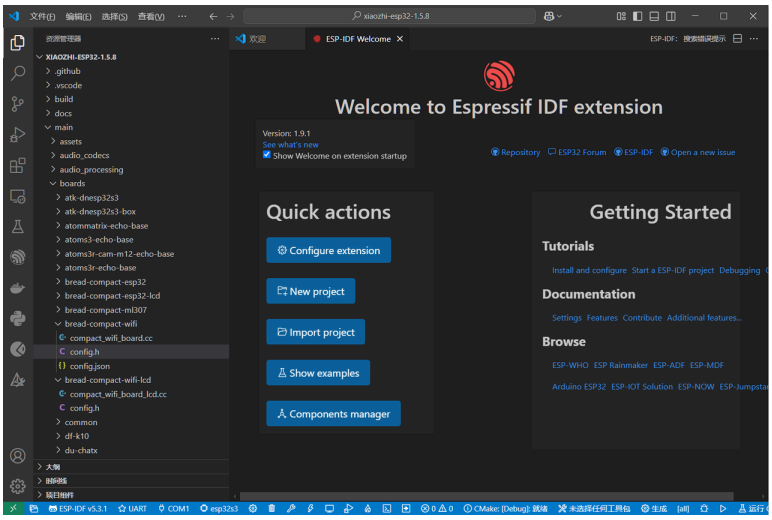
1. Enable real-time conversation mode (AEC) operation with voice interruption

1.1 Use VSCode to modify the configuration SDK of the Xiaozhi AI Terminal Open Source Project

1) Open the Xiaozhi AI Terminal code project in VScode

Based on the directory where the open-source code of Xiaozhi AI Terminal downloaded/cloned by the user is stored on the computer, find the Xiaozhi AI code folder, right-click with the mouse, and open it via Code, as shown in the following figure.

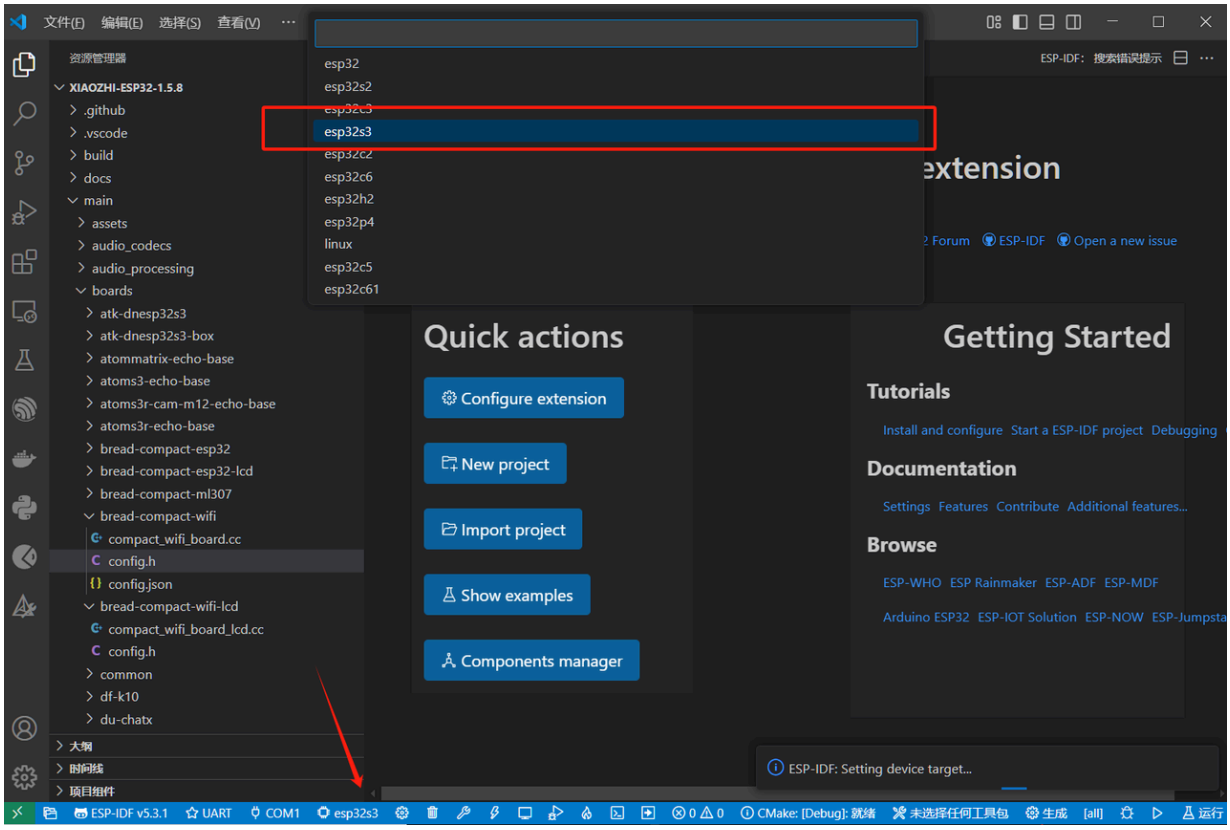


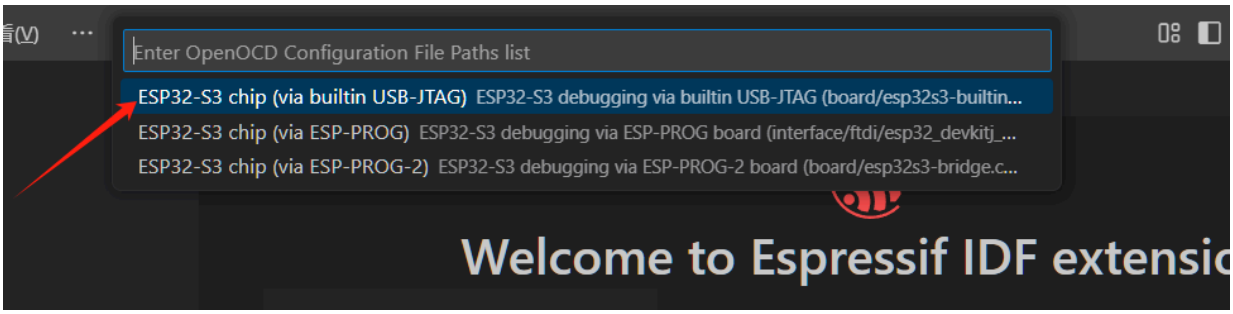


2) Set the target chip to ESP32S3

Select the target chip in the status bar at the bottom of the VScode editor, wait for the successful loading, select esp32s3 in the top lower bar options, and then select the first ESP32-S3 chip (via builtin USB-JTAGE) to complete the template chip setup.

> Note: Currently, only the ESP32S3 development board supports the AEC voice interruption real-time conversation function. First, set the target chip to esp32s3.



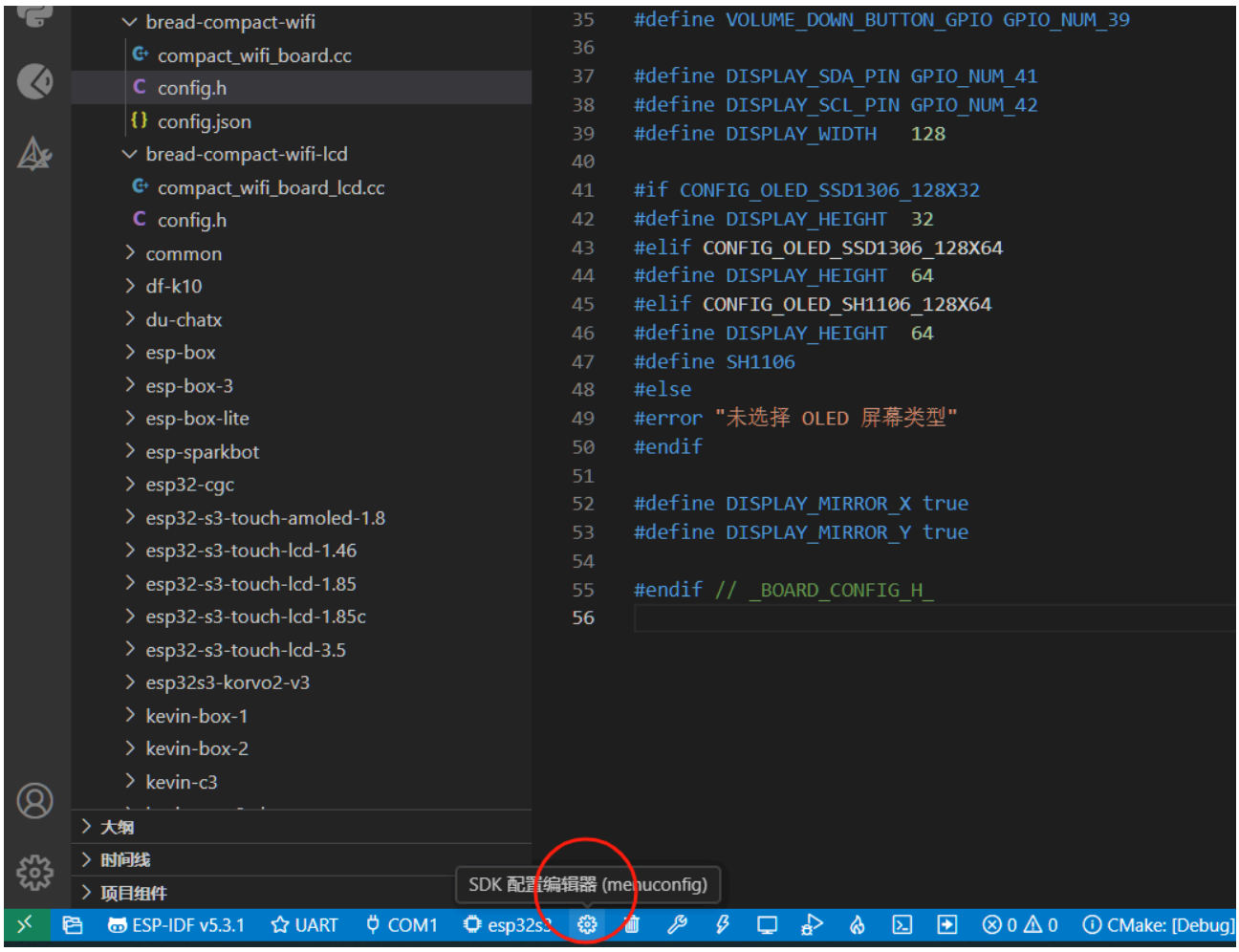


3) Open the SDK to set the development board and AEC options

Click the SDK Configuration Compiler icon in the status bar below, wait for it to load, and then open the SDK configuration, as shown in the following figure.

If you cannot open it, it is generally an issue with the IDF Development Environment configuration in VScode. Please refer to the IDF Development Environment Configuration Tutorial Document. Setting up the ESP IDF 5.3.2 Development Environment and Compiling Xiaozhi on Windows

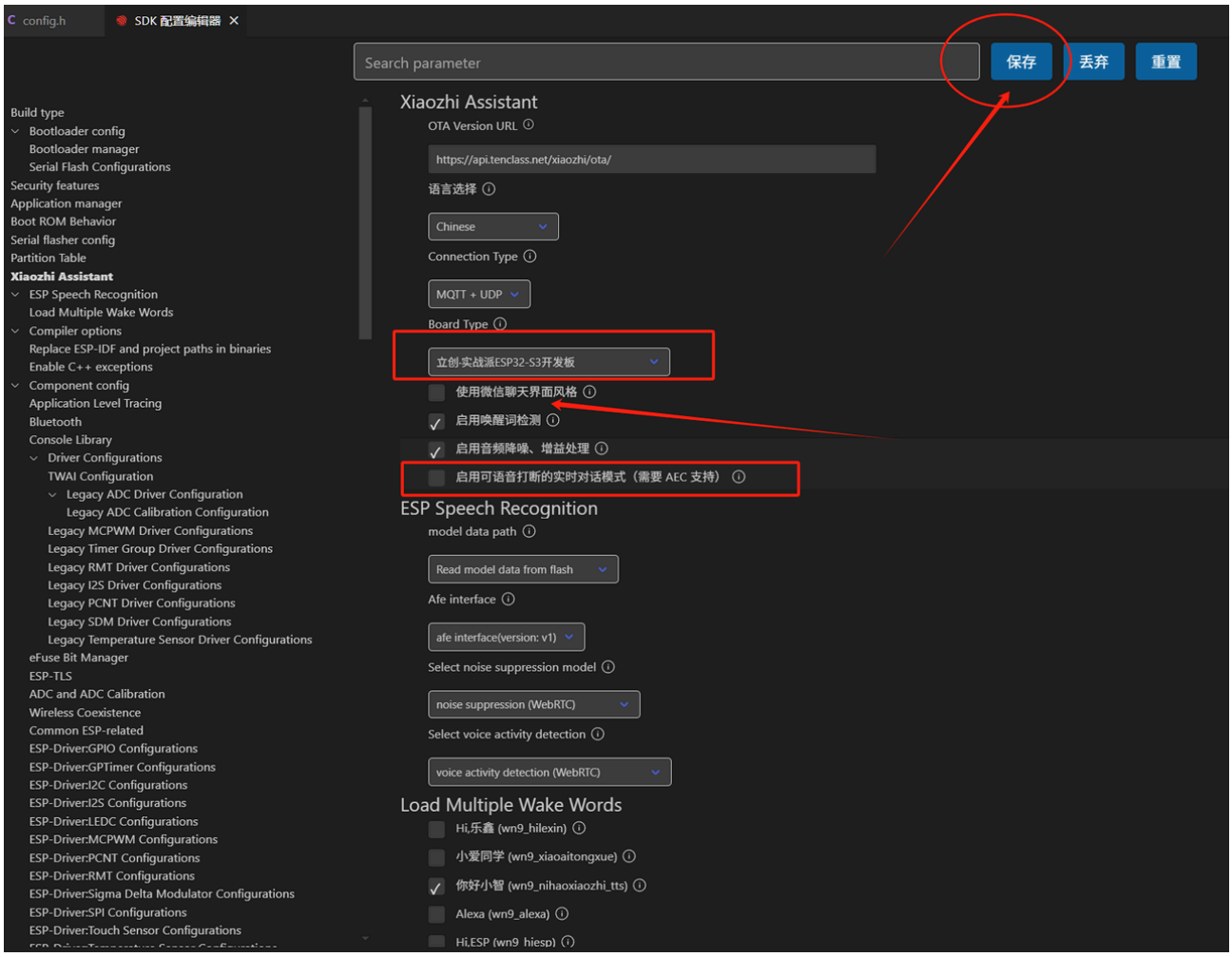
> Note: Since AEC requires the performance of the development board chip and multi-microphone support, currently only some development boards support enabling the AEC option, including the LiChuang·Shizhanpai ESP32-S3 Development Board, ESP BOX, ESP BOX3, and ESP BOX Lite.



Find the Xiaozhi Assistant navigation menu on the left side of the SDK Configuration Editor, and enter the Xiaozhi AI Custom Configuration Options page.

Please select the corresponding development board based on the options listed above, check the configuration to support AEC. Here, we take the Lichuang·Shizhanpai ESP32-S3 development board as an example, and please make the selection according to your own development board specifically.

> Note: Due to high performance requirements, please do not enable both the WeChat chat interface style and AEC support simultaneously



Finally, the most important step is to save the configuration (make sure to save it), recompile the code to generate the firmware, flash it onto the corresponding development board, and after normal operation, the AEC voice interruption real-time conversation function can be implemented.

If there are any omissions or unclear points in the above documents, you can ask questions in the QQ group and discuss them with group members. Thank you!